Gaussian User's Guide

Current as of July 17th, 2008

This document is intended to serve as a guide for running Gaussian jobs quickly on ACEnet resources. The guide contains the same information as the Gaussian wiki (http://wiki.ace-net.ca/index.php/GAUSSIAN). Please forward any suggestions for its improvement to our tech team at support@ace-net.ca.

If you encounter problems that this document can't resolve, please send a detailed description of your issue to our support team at support@ace-net.ca.

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1 Introduction

ACEnet is pleased to be able to provide access to Gaussian on our Placentia2 cluster. Gaussian is an electronic structure program that is commonly used by chemists, chemical engineers, biochemists, physicists and others for research in established and emerging areas of molecular science. More information about Gaussian03 can be found at http://www.gaussian.com

Gaussian jobs can be submitted from both placentia.ace-net.ca and placentia2.ace-net.ca, however interactive jobs can only be run on placentia2.ace-net.ca.

1.1 License Agreement

Before using Gaussian for the first time at ACEnet you must email Bernice Devereaux (bdeverea@mun.ca) with your username, the name of your project leader, and a statement that you agree to the terms and conditions set out in our Gaussian License which can be found at:
http://www.acenet.ca/software/Gaussian_License.pdf.
You will then be added to the list of users able to access the Gaussian binary and the Gaussian nodes.

2 Available Hardware

Gaussian-03 is licensed for use only on ACEnet's Placentia2 cluster. In July 2008 ACEnet added to Placentia2 nineteen nodes specifically intended to support Gaussian-03 usage. These are 4-core nodes made up of two dual-core Opteron 2222 processors running at 3.0GHz. Each node has 64Gb of RAM, and approximately 400Gb of node-local disk for use as fast scratch space.

3 Environment

Before running Gaussian for the first time, you should add the following lines to your login profile.
If your shell is tcsh, add this to ~/.cshrc. (If you don't know what your shell is, use this one.)

```
setenv g03root /usr/local/gaussian
setenv GAUSS_SCRDIR /home/<username>/scratch
source /usr/local/gaussian/g03/bsd/g03.login
```

Be sure to change <username> to your username!
If your shell is bash, add this to your ~/.bashrc:

```
export g03root=/usr/local/gaussian
export GAUSS_SCRDIR=/home/<username>/scratch
source /usr/local/gaussian/g03/bsd/g03.profile
```

3.1 Scratch Space

The environment variable GAUSS_SCRDIR indicates where Gaussian should write its intermediate files, such as the notoriously-large *.rwf. The prescription given above (/home/username/scratch) gives you the most space, but since this points to a network file system it does not give you the greatest speed. If you are willing to manage your scratch file space more carefully then you can consider using

```
setenv GAUSS_SCRDIR /scratch/tmp
```

which points to node-local disk. Much faster, but not as large (36Gb to 150Gb, depending on the node). This approach also suffers from the potential problem that files may be inadvertently left behind by failed runs.

3.2 Automatic Environment Configuration

If you would like to have the default configuration shown above then you can source our pre-built Gaussian environment set up script. All you need to do is to type in

```
source /usr/local/lib/gaussian.sh
```

You can also use the same line in your job scripts (see below).

4 Running Gaussian

Gaussian can be run both interactively on **placentia2.ace-net.ca** and through the N1GE batch scheduler. Users are asked to run all jobs through the N1GE batch scheduler, unless they are testing a configuration and wanting to make sure there are no errors prior to submitting a job to the scheduler. Test jobs should not run longer than 15 minutes.

4.1 Test Jobs

You can run small, short Gaussian calculations interactively (at the terminal prompt). Before any jobs are run, **remember to source the proper Gaussian start file and set your scratch directory properly (see above)**. Jobs can be executed by running

```
g03 < input > output &
```

Users are reminded to keep an eye on interactive jobs and that interactive jobs shouldn't be left to run more than 15 minutes. Jobs longer than this should be run through the N1GE scheduler – see Gaussian Production Jobs.

4.2 Production Jobs

All production class jobs (ie: not testing) must be submitted via N1 Grid Engine. The commands available in N1GE include:

`qsub`	submit a batch job to N1 Grid Engine.
`qstat`	show the status of N1 Grid Engine jobs and queues.
`qdel`	delete a running or waiting job.

More details on using N1GE can be found at http://wiki.ace-net.ca/index.php/Job_control.

4.2.1 Serial Jobs

The following minimal script could be used to run a serial (single-CPU) Gaussian job.

```
#$ -S /bin/csh
#$ -cwd
#$ -j y
#$ -l h_rt=0:30:0
source /usr/local/lib/gaussian.sh
g03 < ./test001.com > test001.log
```

To submit the job, save the above file as "Gaussian_submit.sh" and type:

```
qsub ./Gaussian_submit.sh
```

This will read input from the file "test001.com" and will put the results in the file "test001.log". Scratch files will be stored in /home/<username>/scratch and the job will be run with one process (one CPU). The job is limited to 30 minutes of run time by the line containing "`h_rt=0:30:0`".

4.2.2 Memory

The default memory reserved for your job by N1GE is 2GB. Gaussian handles memory specifications in such a way that this provides 1GB of extra capacity with Gaussian. This is often insufficient, so you may need to know how to modify this setting.

Within the Gaussian input file one might want to change the %MEM setting to use additional memory. *The value for %MEM in your Gaussian input file should be 1GB less than the value specified in the N1GE job submission script.* Conversely, the value specified for h_vmem in your job script should be 1GB greater than the amount specified in the %MEM directive in your Gaussian input file.

The following submit script called `Gaussian_submit_memory.sh` to submit the a serial Gaussian **with additional memory requirements**:

Here is a sample script called Gaussian_submit._memory.sh:

```
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o test_Gaussian_N1GE.out
#$ -l h_vmem=6G
#$ -l h_rt=12:00:00
hostname
date
source /usr/local/lib/gaussian.sh
g03 < ./testmem.com > testmem.out
date
```

NOTE: Inside testmem.com there needs to be a line that says %MEM=5GB (1GB less than the 6GB specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

To submit the job, type:

```
qsub ./Gaussian_submit_memory.sh
```

To determine whether your job is waiting, running, or finished, use the qstat command.

4.2.3 Parallel Jobs

The submission of parallel jobs to N1GE is similar to the submission of serial jobs, except that the job submission script references the parallel environment and the Gaussian input file must include an %NPROCS directive.

"Your mileage may vary" when it comes to the performance of parallel Gaussian work. Typically Gaussian does not scale well beyond 8 processors , and often fewer. This means, running a job on 8 processors will not give you your results in anything like one-eighth the time it would take on one processor. We encourage you not to use large numbers of processors for Gaussian jobs unless and until you are confident that the turnaround time will be worth it.

In the Gaussian input file the %NPROCS directive is required for Gaussian to use additional processors. This number is *in addition to* the one processor already required. So the processor count specified in the N1GE job script must be one greater than the %NPROCS value.

For example, one might create the following submit script called `Gaussian_Parallel_submit.sh` to submit a Gaussian parallel job that needs 5 processors and 6GB of memory *per processor*:

```
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o test_Gaussian_N1GE.out
#$ -l h_vmem=6G
#$ -l h_rt=2:00:00
#$ -pe gaussian 5
hostname
date
source /usr/local/lib/gaussian.sh
g03 < ./testpar.com > testpar.out
date
```

NOTE: inside testpar.com there needs to be a line that says %MEM=5GB (1GB less than the 6GB specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

NOTE: inside testpar.com there needs to be a line that says %NPROCS=4 (1 less than the 5 processors specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

To submit the job, type:

```
qsub ./Gaussian_Parallel_submit.sh
```